Targetless Online Stereo–IMU Extrinsic Calibration with Drift Tracking and Knock Recovery

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Abstract

Fusing a stereo camera with an IMU requires their extrinsic calibration, which is usually measured with a target and an expert, and measured again whenever the mount shifts. This work estimates it continuously from the rig's own motion instead. An error-state Kalman filter in C++17 updates the extrinsic, the IMU biases and the camera–IMU time offset from reprojection residuals of scene points that are triangulated and marginalised on the fly, so no target or map is needed. The extrinsic is modelled as a slow random walk, a χ^2 test on the filter's own corrections detects knocks, the pixel noise is estimated online, and a status report says when the calibration can be trusted. In 50 simulated runs on a figure-eight with roll, pitch and yaw, the filter reached 0.5° / 1 cm from a 5° / 5 cm error in every run (median 6 s) and was within 0.03° / 0.41 cm at 60 s; it tracked a mount drifting 1° over 10 min to within 0.061°, and was back within tolerance 2.6 s (median) after a 1.5° / 2 cm knock. All results are from simulation.

Keywords: extrinsic calibration, visual–inertial sensing, error-state Kalman filter, targetless calibration, change detection.

1 Introduction

Visual–inertial systems need the rigid transform between camera and IMU: three rotation and three lever-arm parameters. The aim here is a calibrator that runs while the rig operates, follows slow drift, recovers from sudden changes, and reports whether its estimate is good enough to use.

2 Method

2.1 Filter and measurement

The 28 error states are the IMU pose, velocity and biases, the extrinsic, the time offset, and a clone of the previous frame's pose. The filter is adapted from the author's Aether6 navigation filter, whose numerics, IMU noise models and Monte Carlo harness it reuses. At each 20 Hz stereo frame, every point seen in the previous and the current frame is triangulated and marginalised by projecting its eight pixel residuals onto the left null space of its Jacobian, leaving five that depend only on the state. The analytic Jacobians, checked numerically, include the time offset. A per-point χ^2 gate rejects wrong matches, and about 500 rows are QR-compressed to 28 before a Joseph-form update.

2.2 Drift and knocks

The extrinsic follows a random walk, so a drifting mount is followed rather than treated as fixed. Once the calibration has converged, a knock shows up as corrections that keep pushing one way: over 2 s, their sum is tested against its predicted covariance (statistic > 40, against χ^2_6). On detection the extrinsic covariance is re-opened to 2° / 2 cm, and the navigation states are loosened too, since they absorbed part of the knock before it was detected.

2.3 Self-tuning and self-assessment

Projected off every state direction, the residual is pure pixel noise, so a robust estimate of its level, taken before the gate, rescales the noise model. A status function reports 2σ bounds along the least-certain directions, declares convergence when both are inside 0.5° / 1 cm, and names the rotation that would observe a poorly observed direction.

3 Results

Each run draws its own scene, true mount, IMU biases, noise and direction of the 5° / 5 cm starting error; the estimator receives only that starting value (Table 1, Fig. 1).

Table 1. Results over 50 runs per scenario (p50 = median).

Scenario (target)	Result
Converge from 5° / 5 cm (< 0.5°, < 1 cm in 60 s, ≥ 95 % of runs)	**50 / 50**; p95 at 60 s 0.025° / 0.34 cm
Drift 0.1°/min for 10 min (error < 0.3°)	**50 / 50**, worst 0.061°; mount assumed fixed: 0.55°
Knock, 1.5° / 2 cm	detected 0.25 s, in tolerance 2.6 s (p50)
8 ms clock offset; 5 % wrong matches; 2× pixel noise	50 / 50 converge in each

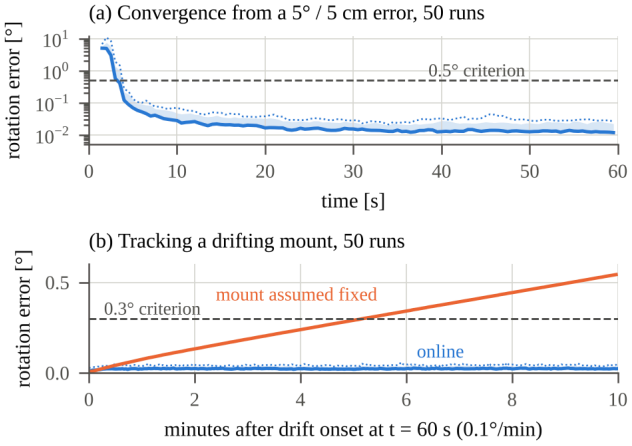


Figure 1. Rotation error, 50 runs: median, p50–p95 band, maximum (dotted). (a) Convergence. (b) From t = 60 s the mount drifts at 0.1°/min; the same filter with the mount assumed fixed falls behind. More in Appendix A.

In 200 runs on a still mount (nominal, clock offset, wrong matches, 2× noise) and through the drift, the status never said *converged* with the true error outside 0.5° / 1 cm. It certified about 10 s after the truth got inside (36 s with 2× noise). The change detector raised no false alarm in 200 runs (11.7 h of data).

4 Limitations and Next Steps

The data are simulated: an ideal pinhole stereo pair, given correspondences and known IMU noise. The motion matters: with a fifth of the roll and pitch and half the yaw, the lever arm reaches 1 cm in only 34 / 50 runs (median 50 s), and the status, correctly, certifies none. The covariance is conservative (NEES 1.9 against 6), which the 2σ status relies on; a short sliding window of poses would make it consistent and add information under gentle motion. Validation on recorded data is next.

Appendix A Convergence and Knock Recovery

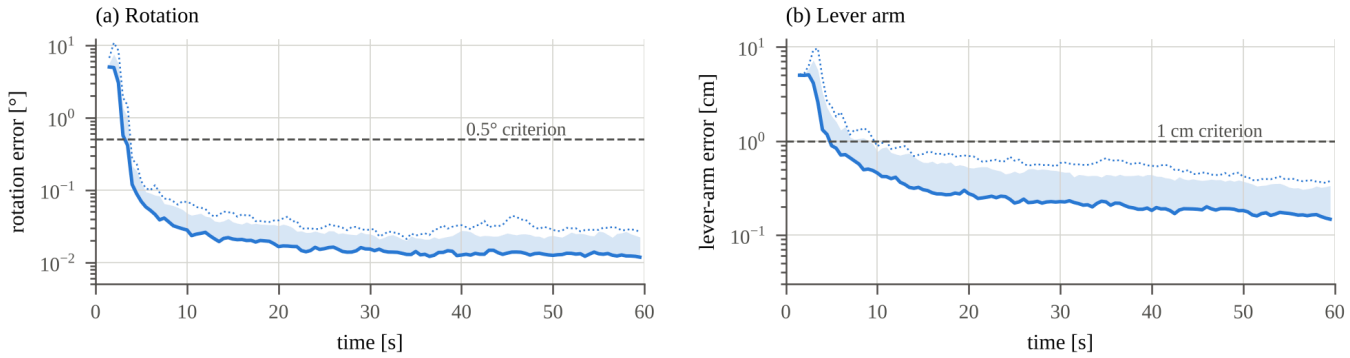


Figure A1. Convergence from a 5° / 5 cm starting error, 50 runs: median (line), p50–p95 band and maximum (dotted). The lever arm is the slower part: at 60 s its error is 0.15 cm (p50) and 0.40 cm in the worst run.

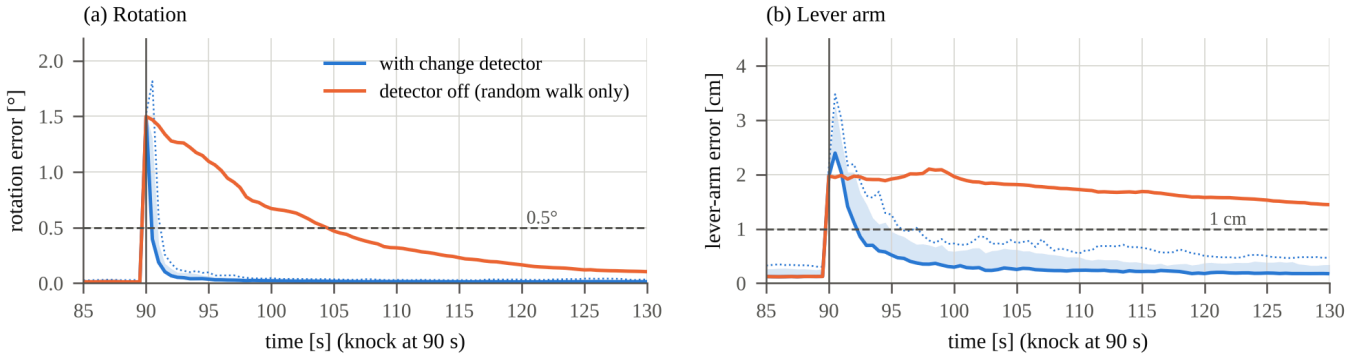


Figure A2. A 1.5° / 2 cm (1.97 cm) knock at 90 s, 50 runs, sampled every 0.5 s. With the change detector (blue: median, p50–p95 band, maximum) the knock is detected after 0.25 s (p50; 0.40 s at most) and the estimate is back in tolerance after 2.6 s (p50). Taken run by run, the lever-arm error peaks at 2.9 cm (median over runs) during re-convergence, and in 8 runs the rotation overshoots the knock by more than 0.1° (2.0° at most); the plotted curves peak lower because these peaks fall between samples. Throughout, the status reports *recalibrating*. Without the detector (orange, median) the random walk recovers the rotation slowly, and the lever arm is still 1.05 cm off at 180 s.

Appendix B Self-Assessment and Sensitivity

A *false claim* is a frame on which the status says *converged* while the true error is outside 0.5° / 1 cm. The 2σ bounds are filter sigmas, not guarantees; the true error exceeded them on 0–0.04 % of the converged frames on a still mount and on 1.3 % during the drift.

Table B1. Self-assessment, 50 runs per scenario. “Converged” is the true error inside 0.5° / 1 cm at 60 s; “status” is what the filter reported.

Scenario	Converged	Status says converged	Status certified (p50)	Truth inside (p50)	False claims
Nominal, clock offset, wrong matches	150 / 150	150 / 150	16.3–17.0 s	6.0–6.4 s	0
Pixel noise 2× configured, estimated	50 / 50	50 / 50	45.6 s	8.1 s	0
Drift, 10 min at 0.1°/min	50 / 50	50 / 50	16.3 s	6.0 s	0
Yaw-only motion	5 / 50	0 / 50	–	–	0
Knock (all before detection)	50 / 50	50 / 50	16.3 s	6.0 s	253
Noise estimation off (model wrong)	33 / 50	49 / 50	22.8 s	29.3 s (33 runs)	15 166
Clock offset ignored (model wrong)	21 / 50	0 / 50	–	–	455

Table B2. Sensitivity checks: same 50 seeds, one setting changed.

Change	Result
Gentler motion: a fifth of the roll and pitch, half the yaw	34 / 50 converge (median 50 s; lever arm 0.88 cm p50 at 60 s); the status certifies none, 0 false claims
Pixel noise configured 10× too low or too high, with or without 5 % wrong matches	50 / 50 converge in all four, unchanged accuracy, 0 false claims; noise estimated as 0.491–0.513 px (10× is the limit of the estimate’s range)
Status at 3σ instead of 2σ	certifies at 34.1 s instead of 16.3 s; never within 60 s with 2× noise
Each observation used once (disjoint frame pairs)	NEES 5.3 (near consistent) but twice the error; the 2σ status makes 240 false claims
5° / 5 cm knock	back in tolerance after 3.6 s (p50); 75 false claims, all before detection

Appendix C Implementation

Estimator and simulator (C++17, Eigen). Adapted from Aether6: its 18-state navigation filter became the 28-state calibrator. Reused from Aether6: rotation algebra, discretisation, the error-state filter numerics (Joseph-form update, multiplicative attitude reset), IMU white-noise and bias random-walk models, and the deterministic multi-threaded Monte Carlo harness. New: a stereo camera with a landmark scene and a true mount that drifts or is knocked; a scripted figure-eight trajectory with roll, pitch and yaw excitation; the reprojection model with analytic Jacobians; the change detector, noise estimate and status; and the scenarios. No ROS.

Verification. 117 Catch2 test cases: Jacobians against central differences, a noise-free run that must stay at the truth, propagation order of accuracy, bit-for-bit reproducibility from a seed, and end-to-end scenario checks. CI builds with GCC and Clang (Release, warnings as errors) and runs a Debug subset with Eigen's assertions on. GCC and Clang (both with libstdc++) produce identical runs. `make eval` regenerates every Monte Carlo result here, and `docs/writeup/make_figures.py` the figures; the development history in Appendix D is documented in `docs/method.md`.

Appendix D Evaluation Protocol and Negative Results

- The estimator never sees the truth; all scoring against it happens outside the filter. Seeds derive from one master seed, so any run can be reproduced exactly.
- The noise-free regression test exposed a propagation bug: the first IMU sample of each step was read by reference after being overwritten, a hidden 2.5 ms time shift that, before the clock offset was estimated, caused a 0.3° systematic error during convergence.
- Without clock-offset estimation, an unmodelled 2.5 ms camera–IMU offset biases the lever arm by about 1 cm (1.07 cm p50) and only 21 / 50 runs converge, so the offset is always estimated.
- The first noise estimate was taken after the gate. With the noise configured four or more times too low, the gate rejected the data that would have corrected it and the covariance collapsed. Measuring before the gate fixed that; wrong correspondences then inflated even the median through the projection, which a coarse pre-screen and a trimmed repeat fixed.
- After a knock, re-opening only the extrinsic let the status re-certify a lever arm still more than 1 cm off (135 frames over 50 runs), and a 1° / 1 cm re-open did the same for a 5° / 5 cm knock (283 frames). Loosening the navigation states and re-opening to 2° / 2 cm removed both.
- Using each observation once makes the covariance nearly consistent but doubles the error (p95 0.045° / 0.73 cm against 0.025° / 0.34 cm), which is why the sliding window in Section 4 is the next step rather than simply dropping half the data.